

JEE MAIN REVISION MASTER: S-BLOCK ELEMENTS OUT OF SYLLABUS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 3 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Standard	S Block Elements — Core Principles & Governing Laws	1 PYQs	Core Principle
Standard	S Block Elements — Standard Calculations & Problem Archetypes	1 PYQs	Quantitative Analysis
Standard	S Block Elements — Applied Cases, Reactions & Traps	1 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
S Block Elements — Core Principles & Governing Laws	$\text{BeO} + 2\text{NH}_3 + 4\text{HF} \xrightarrow{\text{A}} (\text{NH}_4)_2[\text{BeF}_4]$
S Block Elements — Standard Calculations & Problem Archetypes	mol^{-1}
S Block Elements — Applied Cases, Reactions & Traps	$\text{Li}_2\text{O}, \text{N}_2, \text{O}_2, \text{LiNO}_2, \text{NO}_2, \quad \text{LiNO}_3$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 S Block Elements — Core Principles & Governing Laws (Standard Yield — 1 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in S Block Elements.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in S Block Elements.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in S Block Elements PYQs.

Representative Source PYQs:

• [2022 Evening Shift] Q#3 (Numerical): "BeO reacts with HF in presence of ammonia to give [A] which on thermal decomposition produces [B] and ammonium fluoride. Oxidation state of ..." [Ans: 2]

3.2 S Block Elements — Standard Calculations & Problem Archetypes (Standard Yield — 1 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in S Block Elements.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2023 Evening Shift] Q#1 (Numerical): "The number of alkali metal(s), from Li, K, Cs, Rb having ionization enthalpy greater than 400 kJ mol^{-1} and forming stable super oxide..." [Ans: 2]

3.3 S Block Elements — Applied Cases, Reactions & Traps (Standard Yield — 1 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of S Block Elements.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2023 Evening Shift] Q#2 (Numerical): "On heating, LiNO_3 gives how many compounds among the following _____ Li_2O , N_2O , O_2 , LiNO_2 , NO_2 ..." [Ans: 3]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2023 (Online) 31st January Evening Shift (Q#1)

The number of alkali metal(s), from Li, K, Cs, Rb having ionization enthalpy greater than 400 kJ mol^{-1} and forming stable super oxide is _____.

Correct Answer: 2

Solution Analysis: Li does not form superoxide. K, Rb and Cs form stable super oxides but Cs has ionisation enthalpy less than 400 kJ . While K and Rb have ionisation enthalpy greater than 400 kJ mol^{-1} .

Benchmark Problem 2: JEE Main 2022 (Online) 27th June Evening Shift (Q#3)

BeO reacts with HF in presence of ammonia to give [A] which on thermal decomposition produces [B] and ammonium fluoride. Oxidation state of Be in [A] is _____.

Correct Answer: 2

Solution Analysis: $\text{BeO} + 2\text{NH}_3 + 4\text{HF} \rightarrow (\text{NH}_4)_2[\text{BeF}_4]$
 $(\text{NH}_4)_2(\text{BeF}_4) \rightarrow \text{BeF}_2 + 2\text{NH}_4\text{F}$ Oxidation State of Be in (A) is (+2)